

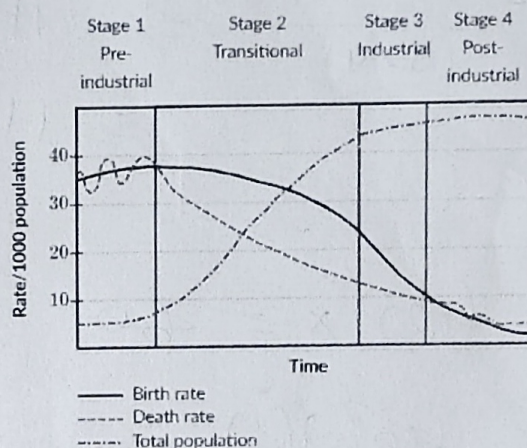
- 1) How was Hubble able to calculate a rough estimate for the formation of the Universe?
- a) By calculating the age of the oldest atoms
  - b) By calculating when hydrogen was first formed
  - c) By calculating the rate at which the Universe is shrinking
  - ☒ d) By calculating the rate at which the Universe is expanding
- 2) Why is the increasing temperature of matter an important step in the lighting up of stars?
- a) It allows for hydrogen to be converted into uranium.
  - b) It allows the star to eject its outer layers of heavy elements.
  - ☒ c) It allows for fusion reactions that release incredible amounts of energy.
  - d) It allows subatomic particles to come together to form atoms.
- 3) What are the four most common elements that make up the Earth's crust?
- a) Oxygen, hydrogen, helium, iron
  - b) Hydrogen and helium
  - c) Oxygen, silicon, helium, iron
  - ☒ d) Oxygen, silicon, aluminum, iron
- 4) Why are some of the planets in our Solar System gaseous planets, and some rocky?
- ☒ a) The planets closest to the Sun had their gases blown off by stellar winds, leaving only their rocks and metals behind. The gas blown further out into the Solar System condensed to make the gassy planets.
  - b) The planets furthest from the Sun had greater concentrations of space debris and dust in their orbits, which is why they formed into rocky planets.
  - c) The planets furthest from the Sun had their gases blown off by stellar winds, leaving only their rocks and metals behind. The planets that are closest to our Sun remained gaseous.
  - d) The planets closest to the Sun had greater concentrations of space debris and dust in their orbits, which is why they formed into rocky planets.
- 5) What percent of atomic matter in the Universe is responsible for creating objects more complex than stars?
- ☒ a) 2%
  - b) 42%
  - c) 74%
  - d) 98%
- 6) Which of the following best describes the state of the Earth in its infancy, 4.5 billion years ago?
- ☒ a) A hot and inhospitable place for life
  - b) A freezing cold and inhospitable place for life
  - c) A hot but hospitable place for life
  - d) A freezing cold but hospitable place for life
- ☒ 7) Around 2.4 billion years ago, the Great Oxidation Event occurred, and many anaerobic respirators died off in an ecological disaster. But it wasn't all bad. What was the major benefit of this event?
- a) The Earth's temperature dropped
  - ☒ b) The formation of the ozone layer
  - c) A chance for mammals to thrive
  - d) The emergence of multicellular organisms
- ☒ 8) What is the result of tectonic plates quickly sliding against each other at sites where strain has built up?
- ☒ a) Volcanic eruptions
  - b) Creation of new plates
  - c) Continental drift
  - ☒ d) Earthquakes
- 8) The fossil record shows that in just the last 600 million years, there have been five instances of sharp decreases in the diversity and abundance of living organisms. What are these events?
- a) Reversals of the Earth's magnetic field
  - b) Droughts
  - c) Great migrations
  - ☒ d) Mass extinctions
- ☒ 9) What was the Cambrian Explosion?
- a) A volcanic eruption that triggered a mass extinction
  - b) A rapid increase in the diversity of life
  - ☒ c) The first emergence of multicellular organisms
  - ☒ d) An asteroid collision that wiped out the dinosaurs



- 1) The demographic transition model hypothesizes that as countries become industrialized, they have predictable patterns of population change. The demographic transition model is shown below.

Which of the following is most likely to explain why a country's birth rate declines during Stage 4?

- a) The country falls into military conflict and war within its borders.
- ☒ b) The country improves food and nutrition access.
- ☒ c) The country improves its public education system.
- d) The country experiences a period of economic depression.



- 2) Japan has a population of 120 Million and growth rate of -0.5% per year and its population is shrinking. What will be its population 10 years from now? Answer: 117187500 people
- 3) How long will the population take to shrink to half Answer: 5.64 years  
139
- 4) Why might historians describe agriculture as an energy "bonanza" for humans?
- ☒ a) Because agriculture gave humans the ability to grow a surplus of food, which gave them much more energy
  - b) Because the climate had been warming up, releasing a greater amount of the Sun's energy
  - ☒ c) Because agriculture allowed humans to use a greater amount of the Sun's energy
  - d) Because agriculture allowed humans to settle down for the first time
- 5) The modern revolution came about not only because of the globalization of goods and ideas but also due to the utilization of new energy resources. What was this new energy source?
- a) Hydroelectric power
  - ☒ b) Fossil fuels
  - ☒ c) Nuclear energy
  - d) Firewood
- 6) How did the Industrial Revolution and early farming impact the environment differently?
- a) The Industrial Revolution cleared forests, while farming did not.
  - b) Farming increased average global temperatures, while the Industrial Revolution did not.
  - c) Farming burned natural resources, while the Industrial Revolution did not.
  - ☒ d) The Industrial Revolution dramatically changed the Earth's atmosphere, while farming did not.